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Submitted to

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(CoAL)

(La2 = 2)

(Tasks)

1) Describe the function of each.

•) **Accumulator Register (A, Ax, EAx, RAx)**

It is used for arithmetic, logic, and input/output operations; often stores intermediate results.

•) **Base Register (B, Bx, EBx, RBx)**

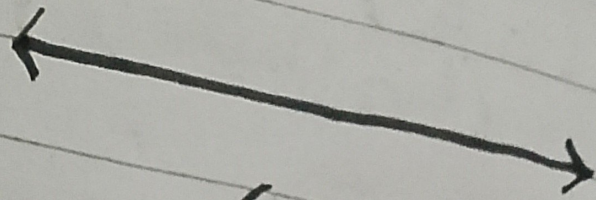
It is commonly used to hold memory addresses (Base pointers) for data access.

•) **Counter Register (C, Cx, ECx, RCx)**

It is used as a loop counter, shift/rotate counter, and in string operations.

•) **Data Register (D, Dx, EDx, RDX)**

Stores data for arithmetic, Input/output operations and sometimes extended



(2)

Describe the role of index registers in source and destination operations. Provide an example using SI and DI is an assembly instruction

Role of Index Registers:-

- SI (Source Index)
Points to the source data in memory (used in string and array operations)

Example:-

mov

ab ← 2
destination source

- DI (Destination Index)

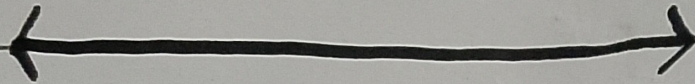
Points to the destination in memory where data will be written

Date: _____

Day: _____

Example:-

mov 2 ab
 source destination



(3)

Briefly describe the Functions of these bits

•) Overflow Flag (OF)

(1) set when signed arithmetic results exceed the representable range.

Example:-

127 + 1 adding 127 + 1
exceeds signed
8-bit range (-128 to +127)

•) Zero Flag (ZF)

(1) set when the result of an operation is zero

Example

5 - 5 = 0 → set (1) result

Date: _____

Day: _____

•) Carry Flag (CF)

(1) set when an unsigned operation produces a carry/borrow out of the most significant bit

Example

$255 + 1$ (produces carry out of the most significant bit)

